MIL-T-19247A(SHIPS)
17 February 1961
SUPERSEDING
MIL-T-19247(SHIPS)
14 September 1956

MILITARY SPECIFICATION

TANK, LIQUID STORAGE, NONMETALLIC,

MARINE OIL AND WATER SERVICE

1. SCOPE

1.1 This specification covers non-magnetic fibrous glass reinforced laminated tanks to be used for storage of fuel, lubricating and hydraulic oils, and drinking water onboard naval ships under cold service temperature not to exceed 125° Fahrenheit (F.).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-R-7575 - Resin, Polyester, Low pressure, Laminating.
MIL-W-16791 - Detergent, Nonionic.
MIL-W-17549 - Plastic Laminates, Fibrous Glass Reinforced, Marine Structural.
MIL-R-21607 - Resins, Polyester, Low Pressure Laminating, Fire Resistant.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Materials. - Reinforced plastic laminated material shall be in accordance with Specification MIL-P-17549 and as specified herein. Other materials not definitely specified shall be of good commercial quality entirely suitable for the purpose intended.

3.1.1 Plastic laminate. - The plastic laminate furnished under this specification shall meet the requirements for grade W of Specification MIL-P-17549.

3.1.2 Resin. - For tanks intended for storage of fuel, lubricating oil or inflammable hydraulic oil, a fire resistant resin in accordance with class A of Specification MIL-R-21607 shall be used; otherwise general purpose resin conforming to grade A, class D of Specification MIL-R-7575 shall be used.

3.1.3 Chemical resistance of laminate. - The tank laminate shall be resistant to the liquid to be stored and shall not have a deleterious effect upon the liquid.

3.1.3.1 For oil storage tanks, laminate tested as specified in 4.3.2 shall meet the (flexural) "wet" strength requirements of Specification MIL-P-17549. Laminate shall show no evidence of softening, swelling, delamination or other attack by the liquid.

3.1.4 Toxicity. - The material of tanks intended for drinking water storage, shall have been approved as safe for use in drinking water systems (see 4.4).

3.1.5 Metallic components. - The tank assembly shall be constructed of nonmagnetic material (material which has a maximum magnetic permeability not greater than 2 after fabrication).

3.1.6 Pigment. - To facilitate visual examination, the resin shall not be pigmented and the laminate shall be translucent.

3.1.7 When any specification which forms a part of this specification requires that the product be subjected to and pass qualification tests, only products which are listed on the applicable Qualified Products List on the date of invitation for bids or which may be added to that Qualified Products

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List subsequent to that date shall be utilized in the construction of equipment specified to be in accordance with this specification. In the event no Qualified Products List has been issued, the contractor shall request instructions as to what testing will be required to determine whether the product meets the requirements of the specification.

3.2 Painting. - When painting is specified (see 6.1), it shall not be done until the tank has passed visual examination as specified (see 4.2.1).

3.3 Performance requirements. -

3.3.1 Hydrostatic pressure. - Unless otherwise specified (see 6.1), the tank shall show no evidence of leakage, weeping, permanent set, or structural failure when tested as specified in 4.3.1 at a hydrostatic test pressure of 1-1/2 times the working pressure. Deflection of the tank shall not exceed 1/100 of the shortest span of the largest tank panel. In any case, deflection shall not exceed 1/2 inch.

3.4 Tank construction. -

3.4.1 The tank may be of one piece contruction or broken down into sections or pieces, if necessary, to facilitate installation or shipping. Any sealing materials used shall be impervious to and chemically resistant to the liquid to be stored.

3.4.2 Tank shell, internal stiffening and swash plates. - Tank shell, internal stiffening, and swash plates shall be designed to limit deflection, as required in 3.3.1, and provide a minimum safety factor of 4:1 based on working pressure.

3.4.3 Baffles. - Baffles or swash plates shall be constructed within tanks, as required (see 6.1) or necessary, to dampen the dynamic action of the tank fluid.

3.4.3.1 Baffle arrangement. - The baffle arrangement shall be such as to provide for complete access to the inside of the tank by means of lightening and access holes. They shall not interfere with the required manholes, tank connections, and internal piping. Limber holes at the bottom and air holes at the top shall be provided in all baffles.

3.4.4 Lifting pads. - Lifting pads or some suitable means for handling shall be provided on all tanks having greater than 750 gallon capacity.

3.4.5 Tank openings. - All tank openings shall be reinforced in such a manner as to maintain the strength of the original uncut panel. The location of the tank openings shall be as specified (see 6.1).

3.4.6 Pipe connections. - Typical metallic pipe connections to plastic tanks shall be as shown on figure 1.

3.4.7 Drainage and venting. - Provisions shall be made for complete and rapid drainage and venting of the tank and each compartment within the tank.

3.5 Weight, dimensions and tolerances. - Weight, dimensions and tolerances shall be as specified (see 6.1).

3.6 Tank marking. - The following information shall be indelibly marked on each tank:

(a) Capacity and fluid to be stored.
(b) Working pressure.
(c) Manufacturer, contract number and date of manufacture.
(d) "MIL-T-19247".

3.7 Drawings. -

3.7.1 Preliminary drawings. - Within 30 days after award of contract, the contractor shall submit to the bureau or activity concerned preliminary drawings that are sufficient to permit evaluation of the design and approval of materials. These drawings shall be detail assembly drawings and shall show the following:

(a) Assectional assembly of the tank and details of means for making piping connections.
(b) Bill of material, listing specifications, grade and condition, or other data adequate to identify the materials proposed.
(c) Dimensions - Overall, accessibility space including disassembly clearances when applicable, and all dimensions pertinent to installation.
(d) Notes, design and performance characteristics, weight, any limitations on installation.

3.7.1.1 If a contractor proposes to furnish equipment identical to that which previously has been supplied and for which drawings have been previously approved, resubmission of drawings for approval will not be required. References to the approved drawing (where required by an activity not having the drawing on record) will be acceptable.

3.8 Workmanship. - All finished tanks shall be of good workmanship and uniform quality, be free from loose fibers, indentations, cracks, blisters, porosity and other defects which, due to their nature, degree or extent, detrimentally affect the strength and serviceability of the tanks.

3.8.1 Interior surfaces. - Particular care shall be taken with all interior tank surfaces. The surfaces of laminates shall be resin smooth, non-porous and shall show no evidence of exposed, loose, projecting, or lightly bonded fibers. All holes shall be carefully

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cut to prevent delamination. Where possible and practicable, cut edges of laminates shall be sealed by coating with a suitable resin formulation. Inside surfaces of tanks shall be absolutely clean and free from all foreign matter such as parting compound, loose overlay, glass fibers, dirt and dust.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspection set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Acceptance inspection. -

4.2.1 Visual examination. - Each tank shall be visually examined and measured to verify conformance with all of the requirements of the contract or order and of this specification which do not involve tests. Any unit which is defective in any aspect shall be rejected. Examination shall be conducted as specified in table I.

Table I - Classification of defects in accordance with Standard MIL-STD-105.

Categories	Defects
Critical:	
1	Toxicity (for fresh water storage) nonconforming.
Major:	
101	Not size, capacity specified in ordering data.
102	Evidence of unauthorized materials or repairs.
103	Parts not new; evidence of reclaimed, salvaged, used parts.
104	Parts missing, broken, distorted, cracked, blistered, porous; not free of loose fibers.
105	Laminate not clear; will not permit visual examination.
106	Tank shell, stiffening, baffles, lightening poles not located or dimensioned in accordance with approved process description.
107	Bolts, and bolt holes not diameters within tolerances required by approved process description.
108	Tank opening reinforcements, removable covers not as required by approved process description.
109	Metallic pipe connections locations, size, type specified; tank and compartment drainage not complete or rapid.
110	Tank inside surfaces not clean; not free of parting compound, glass fibers, loose projecting or weakly bonded fibers, other foreign matter.
111	Cut edges not sealed as specified; evidence of de-lamination around cut holes.
112	Tank, interior not treated to prevent glass fibers working out of surface where required by description.
113	Products not listed in qualified products list used without authorization.
114	Tanks not free of loose fibers; not free of indentations, cracks, blisters and porosity.
Minor:	
201	Instructions not supplied; not quantity required; data not complete.

4.2.2 Hydrostatic tests. - Each assembled tank shall be tested in accordance with 4.3.1 to verify conformance with the requirement of 3.3.1. Any tank which fails to meet the specification requirements shall be rejected.

4.2.3 Tests of reinforced plastic laminate material. -

4.2.3.1 Lot. - For the purposes of 4.2.3.3, all tanks manufactured under the same conditions and presented for inspection at one time shall be considered a lot.

4.2.3.2 Sampling for properties of plastic laminate. - Test specimens from a random sample of tanks selected in accordance with table II shall be

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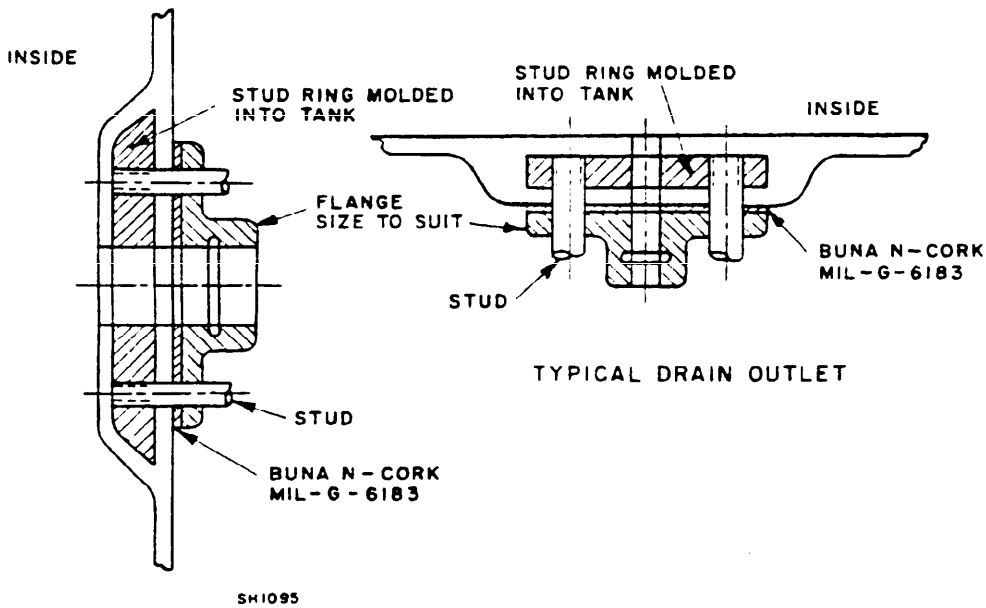


Figure 1 - Pipe connections.

tested as specified in 4.2.3.3. If any sample fails in any test the tank which it represents shall be rejected, and if the number of such rejected tanks exceeds the acceptance number of table II for the sample representing the lot, the entire lot shall be rejected.

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Table II - Sampling of tanks for tests of the material

Number of tanks in lot	Number of tanks in sample	Acceptance number (defective)	Rejection number (defective)
3 and under	1	0	1
4 to 7	2	0	1
8 to 15	3	0	1
16 to 25	5	1	2
26 to 40	7	1	2
41 to 65	10	2	3
66 and over	15	3	4

4.2.3.3 Test specimen samples. - Test specimen samples of the material shall be subjected to the tests specified in 4.3.2. Five specimens are required. The test specimen samples shall be obtained from sections of the tank shell such as manhole, hand hole, or pipe connection openings. Material specimens shall not be taken from tank shell areas which would require restoration repair and thus destroy the integrity of the tank shell. In the event that it is not practicable to obtain suitable test specimens from the tank, the contractor shall propose an alternate procedure by which he will provide suitable specimens on which conformity to the property specifications can be demonstrated.

4.3 Test procedures. -

4.3.1 Hydrostatic pressure tests. - The test liquid used shall be fresh water for fresh water storage tanks. Kerosene, oil, or a mixture of water and 1/4 of 1 percent of weight of wetting agent in accordance with type 1 of Specification MIL-W-16791, shall be used for oil storage tanks. The temperature of the liquid in the tank shall be maintained above the local dew point in order to prevent condensation on the external tank surface. The tank should be filled and the maximum pressure required by the contract (see 3.3.1 and 6.1) or order shall be applied by means of a riser pipe elevated above the top of the tanks. The applied pressure shall be measured by means of a pressure gage or other suitable device installed in such a manner as to indicate the total hydrostatic pressure at the bottom of the tank. Care shall be taken to assure that all air has been bled from the tank. The tank shall be supported during the test in a manner simulating the support that will exist in the vessel, but in such a way that all joints and fittings will be accessible and visible. Blanks shall be fitted to the tank openings as required. During the test, the tank shall be measured to determine compliance with the required deflection limits. The deflection measurements shall be made from a reference frame constructed about the tank in such a manner as to remain fixed with respect to any movement of the tank or its supports. Tanks shall be cleaned free of test liquid prior to shipment. Test pressure shall be applied for a minimum period of 24 hours.

4.3.2 Properties of plastic laminate. - Properties of plastic laminates shall meet the requirements of 3.1.1 through 3.1.3.

4.3.2.1 Chemical resistance of laminate. - Laminate sample specimens, (see 4.2.3.3) shall be immersed in the type of liquid to be stored in the tank for a period of 168 hours at a temperature of 125° F.

4.4 Toxicity. - For tanks intended for drinking water service the plastic laminate materials, coatings and sealants used must have been evaluated and found acceptable by the National Sanitation Foundation (School of Public Health, University of Michigan, Ann Harbor, Michigan), or by another impartial agency as approved by the Government using equivalent testing procedures and standards.

4.5 Possible test failures are defined as follows:

(a) Hydrostatic test: (specified test condition)
  (1) Evidence of leakage, weeping.
  (2) Deflection exceeds specified limits.
  (3) Structure fails.
  (4) Evidence of permanent deformation.
  (5) When pressure is cycled as prescribed tank fails to meet test requirements.

(b) Chemical tests: (specified test conditions)
  (1) Evidence that laminate fails to meet specified "wet" strength tests. (as required).
  (2) Evidence of softening, swelling, deterioration caused by contained liquid.
  (3) Cure, void percentage and resin content not in accordance with approved process.

(c) Permeability test:
  (1) Metallic parts not within specified magnetic permeability limits.

(d) Flammability test: (specified test conditions)
  (1) Laminate specimens do not meet specified flammability limitations.

(e) Mechanical properties:
  (1) Laminates do not meet the required mechanical properties.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. - Tanks shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. - Tanks shall be packed in a manner which will insure carrier acceptance and safe delivery at destination. Shipping container or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.3 Marking. - In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Tank overall dimensions, tolerances and configuration (and maximum weight where required) (see 3.4.3 and 3.5).
(c) Required volumetric capacity of each tank.
(d) Nature of liquid to be stored.
(e) Whether painting is required (see 3.2).
(f) Working and hydrostatic test pressure required (see 3.3.1 and 4.3.1).
(g) Location of openings and pipe connections (see 3.4.5 and 3.4.6).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell, any patented invention that may in any way be related thereto.

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